

exp 9.sci (C:\Users\pach\Documents\exp 9.sci) - SciNotes

exp 6.sci exp 7.sci exp 8.sci exp 9.sci

```
1 clear;
2 clear;
3
4 // Given Values
5 theta_1 = 45; // degrees
6 theta_2 = 70; // degrees
7 medium_1 = 4; // Relative permittivity (er1)
8
9 // Calculation
10 tan_theta = tand(theta_1) / tand(theta_2);
11
12 medium_2 = medium_1 / tan_theta;
13
14 // Display Results
15 disp('Angle 1 (theta1) = ' + string(theta_1) + ' degrees');
16 disp('Angle 2 (theta2) = ' + string(theta_2) + ' degrees');
17 disp('Relative Permittivity (er1) = ' + string(medium_1));
18
19 disp('tan(theta1)/tan(theta2) = ');
20 disp(tan_theta);
21
22 disp('Relative Permittivity (er2) = ');
23 disp(medium_2);
24
```

SciLab 2025.1.0 Console

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SciLab 2025.1.0 Console

```
"Angle 1 (theta1) = 45 degrees"
"Angle 2 (theta2) = 70 degrees"
"Relative Permittivity (er1) = 4"
"tan(theta1)/tan(theta2) = "
0.3639702
"Relative Permittivity (er2) = "
10.989910
```

--> |

$$\underline{\underline{\theta_{2p} = 9}}$$

$$\theta_1 = 45^\circ$$

$$\theta_2 = 70^\circ$$

$$e_n = 4$$

calculate tangent

$$\tan 45^\circ = 1$$

$$\tan 70^\circ = 2.7475$$

calculate ratio

$$\tan \theta = \frac{1}{2.7475}$$

$$\tan \theta = 0.36397$$

$$\Sigma \pi_2$$

$$\Sigma \pi_2 = \frac{4}{0.36397}$$

$$\boxed{\Sigma \pi_2 = 10.99}$$